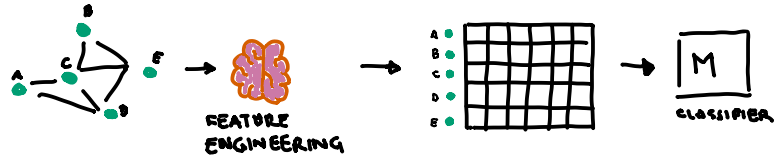


TRADITIONAL APPROACHES FOR ML ON GRAPHS

before the advent of DL approaches

approaches for **classification** use the standard ML paradigm :

- extract statistics / feature (domain knowledge / heuristic)
- use features as input to standard classifier



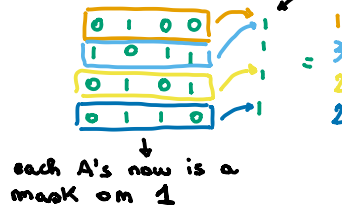
NODE LEVEL STATISTICS → which statistics / properties to use to **CHARACTERIZE NODES**?

useful to DIFFERENTIATE NODES

a) **NODE DEGREE** • given $G=(V,E)$ undirected, the degree $K(u)$ of u is the number of nodes adjacent to u , or $K(u) = |N(u)|$

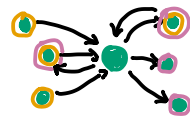
in matrix formulation $K(u) = \sum_{v \in V} A_{u,v} = \sum_{v \in N(u)} A_{u,v} = (A \cdot \mathbf{1})_u$

$N(u) = \{v \in V \mid \{u,v\} \in E(G)\}$
NEIGHBORHOOD of u

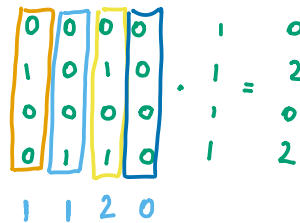


• if G is directed

- IN-NEIGHBORHOOD of u or u 's PREDECESSOR $N^-(u) = \{v \in V \mid (v,u) \in E(G)\} \rightarrow$ IN-DEGREE $K^-(u) = |N^-(u)|$
- OUT-NEIGHBORHOOD of u or u 's SUCCESSOR $N^+(u) = \{v \in V \mid (u,v) \in E(G)\} \rightarrow$ OUT-DEGREE $K^+(u) = |N^+(u)|$



in matrix formulation



$K^+(u) = \sum_{v \in N^+(u)} A_{u,v} = \sum_{v \in V} A_{u,v} = (A \cdot \mathbf{1})_u$

$K^-(u) = \sum_{v \in N^-(u)} A_{v,u} = \sum_{v \in N^-(u)} A_{v,u}^T = (A^T \cdot \mathbf{1})_u$

• degree definitions extensible to weighted graph (summed $w: E \rightarrow \mathbb{R}$)

OUT-STRENGTH

$s^+(u) = \sum_{v \in V} w(u,v) A_{u,v} = (W \cdot \mathbf{1})_u$

$s^-(u) = \sum_{v \in V} w(v,u) A_{v,u} = (W^T \cdot \mathbf{1})_u$
IN-STRENGTH

NODE CENTRALITY → degree is one of the available measures to assess the importance of a node

in degree the importance is only $\propto$ to the size of $N(u)$ but not all $v \in N(u)$ are equally important

EIGENVECTOR CENTRALITY: include the importance / centrality of u 's neighbors
proportional to the sum of importance

start from the basic idea on undir graph $G=(V,E)$ and NOT BIPARTITE

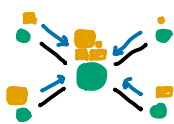
$$e_u = \sum_{v \in V} A_{u,v} \cdot e_v$$

$$\Leftrightarrow e = Ae$$

if we need formula following the → direction

$$e^{t+1} = Ae^t \Rightarrow e^{t+1} = A \cdot Ae^{t-1} = A^2 e^{t-1} = A^3 e^{t-2} = \dots = A^{t+1} e^0$$

initial guess for centrality



property of A : G is undir $\Leftrightarrow A = A^T$
by def $\forall_{u,v} A_{u,v} \geq 0$
 G is connected

A is symmetric
 A is not-negative
 A is irreducible: $\exists$ una permutation P t.c.
 $PAP^T = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}$

a) SPECTRAL THEOREM FOR REAL SYMMETRIC MATRIX

if $A \in \mathbb{R}^{m \times m}$, $A^T = A$ exist

- orthogonal matrix $Q \rightarrow$ column some base orthonormal di $\mathbb{R}^m$
- real diagonal matrix Λ s.t. $\langle v_i, v_j \rangle = 0$ $\|v_i\| = 1$

$$A = Q \Lambda Q^T$$

where $Q^T Q = Q Q^T = I$

$$A = \text{diag}(\lambda_1, \dots, \lambda_m)$$

eigenvalues of A

a) all real

b) eigenvectors for distinct eigenvalues are orthogonal

b) PERRON-FROBENIUS THEOREM

Let $A \in \mathbb{R}^{m \times m}$, and not-negative $A_{ij} \geq 0 \forall i,j$ and irreducible then

- the spectral radius $\rho(A) = \max \{|\lambda| : \lambda \in \sigma(A)\}$ is a positive real eigenvalue of A $\rho(A) > 0$
- $\exists$ right eigenvector for $\rho(A)$ $Ax = \rho(A)x$ $x > 0$ **b2**
- $\exists$ left " " " $y^T A = \rho(A)y^T$ $y > 0$
- the algebraic multiplicity of $\rho(A)$ is 1
- $\forall \lambda \in \sigma(A) \quad |\lambda| \leq \rho(A)$ **b1**

By a) we can decompose $e^0 = c_1 v_1 + \dots + c_m v_m$ and $A v_i = \lambda_i v_i$

$$Ae^0 = A \sum_{i=1}^m c_i v_i = \sum_{i=1}^m \underbrace{A v_i}_{\text{by linearity}} c_i = \sum_{i=1}^m \lambda_i v_i c_i \quad + \text{ inclusion of } \lambda_i \text{ as } A^t e^0 = \sum_{i=1}^m c_i \lambda_i^t v_i$$

$$\Downarrow$$

$$A^{t+1} e^0 = A \cdot A^t e^0 = A \sum_{i=1}^m c_i \lambda_i^t v_i = \sum_{i=1}^m A v_i \lambda_i^t c_i = \sum_{i=1}^m \lambda_i^{t+1} v_i c_i$$

$$\Downarrow$$

$$e^t = \sum_{i=1}^m \lambda_i^t v_i c_i$$

using a stronger condition w.r.t b1 we have $\rho(A) = \lambda_1 > |\lambda_2| \geq \dots \geq |\lambda_m|$ because G is not bipartite

$$e^t = \lambda_1^t \left(c_1 v_1 + \sum_{i=2}^m c_i \left(\frac{\lambda_i}{\lambda_1} \right)^t v_i \right)$$

$\forall i \geq 2 \quad \left| \frac{\lambda_i}{\lambda_1} \right| < 1 \rightarrow \lim_{t \rightarrow \infty} \left(\frac{\lambda_i}{\lambda_1} \right)^t = 0$

for large t : $e^t \approx \lambda_1^t c_1 v_1 \rightarrow$ we normalize $\frac{e^t}{\|e^t\|} = \frac{\lambda_1^t c_1 v_1}{\lambda_1^t \|c_1 v_1\|} = \frac{c_1 v_1}{\|c_1 v_1\|} = \frac{c_1 v_1}{|c_1| \|v_1\|} = \frac{c_1 v_1}{|c_1|} = \text{sign}(c_1) v_1$

by b2 $v_1 > 0$
 $\uparrow$
 $\downarrow c_1 \neq 0$
 $e^0 \geq 0, e^0 \neq 0$

the limiting vector of centralities is proportional to the leading eigenvector of A

$$Ae = \lambda_1 e \rightarrow e_i = \frac{1}{\lambda_1} \sum_{j=1}^m A_{ij} e_j \quad \text{EIGENVECTOR CENTRALITY}$$

to compute eigenvector centrality, power method is a possible solution

it iteratively applies the adj matrix A to the centrality vector e

- start from a initial vector $e^0 \neq 0$, usually $e^0 = 1$

- while True

$$e^t = A e^{t-1}$$

$$e^t = \frac{e^t}{\|e^t\|} \rightarrow \text{normalization to avoid } \|e\| \text{ increases at each iteration}$$

$$\text{if } \|e^t - e^{t-1}\| < \epsilon \rightarrow \text{stop condition break}$$

at $t=1$, $e^1 = A1$ contains nodes' degree $\rightarrow$ at $t=2$ each node receives the degree of its neighbors

there are many more measures of centrality $\rightarrow$ BETWEENNESS CENTRALITY $\rightarrow$ node is important if it lies on many shortest paths between other nodes
 CLOSENESS CENTRALITY

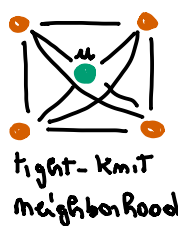
node is important if it has small shortest paths to all other nodes

$$b(\mu) = \sum_{s, d, \mu \in V, s \neq d \neq \mu} \frac{\# \text{ shortest paths } (s, d) \text{ containing } \mu}{\# \text{ shortest paths } (s, d)}$$

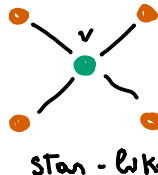
$$C(\mu) = \frac{1}{\sum_{\substack{v \in V \\ v \neq \mu}} |\text{shortest path } (\mu, v)|}$$

CLUSTERING COEFFICIENT the structure around a node may help in differentiate nodes as well as degree and centralities

start from its density



$\neq$



structural diversity can be measured by the
 $\Rightarrow$ CLUSTERING COEFFICIENT = proportion of triangles in the node's neighborhood

given $G = (V, E)$ and $S \subseteq V$ the node-induced subgraph of G on S is the graph $G[S] = (S, E_S)$ exactly the nodes in S

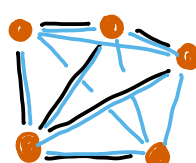
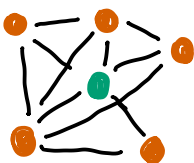
where

$$E_S = \{ \{u, v\} \in E \mid u, v \in S \}$$

$\downarrow$
 all links in E whose endpoints belong to S

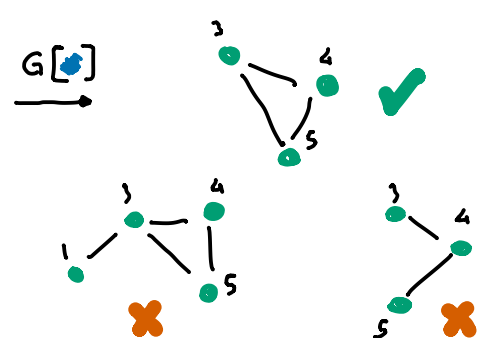
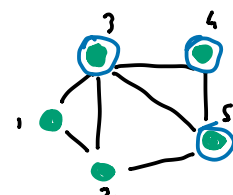
for a node μ , its EGO-NETWORK is a subgraph centered around μ

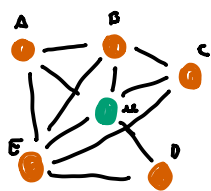
$$cc(\mu) = \frac{|E(G[N(\mu)])|}{\binom{k_\mu}{2}} \in [0, 1] \quad \left\{ \begin{array}{l} \text{a) ego-network excluding the ego} \\ G[N(\mu)] \end{array} \right.$$



$$cc(\mu) = \frac{6}{10} = 0.6$$

$$cc(\mu) = \frac{\# \text{ triangle containing } \mu}{\# \text{ triads centered in } v} \quad \left\{ \begin{array}{l} \text{b) ego-network including the ego} \\ G[\{\mu\} \cup N(\mu)] \end{array} \right.$$





$$\left. \begin{array}{ll} A-u-B & B-u-D \\ A-u-C & B-u-E \\ A-u-D & C-u-D \\ A-u-E & C-u-E \\ B-u-C & D-u-E \end{array} \right\} \binom{K_{u,u}}{2}$$

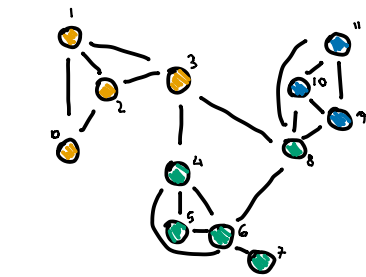
centered in u
 $\uparrow$
 $\Rightarrow$ counting a specific subgraph
in a graph containing u



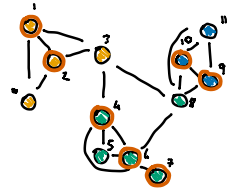
among these triads how many are closed? = TRIANGLE

the above definition holds for undir graph $\rightarrow$ extensions available for directed graphs ("Networks: An Introduction", Mark Newman, 2018)

e.g. node classification in Coxa by ML with feature engineering

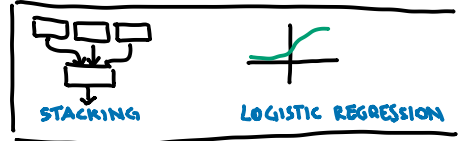


$\Rightarrow$ NODE MASKING
(train, val, test split)



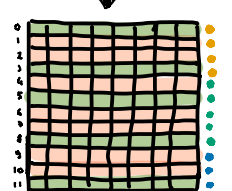
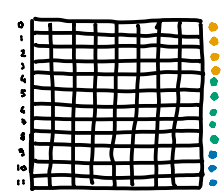
train 0 1 1 0 1 0 1 1 0 1 1 0

MODEL SELECTION

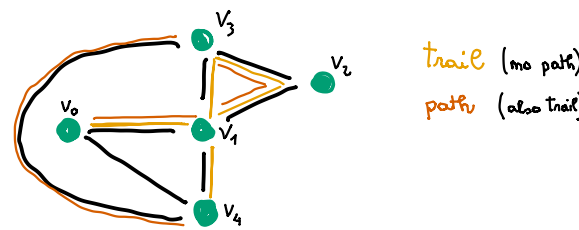


2 classifiers

training on red rows
testing/inference on green rows



length of the walk



PATH is a walk where no node is repeated, except v_0 and v_k .

↳ path from u to v is the sequence $u = v_0, \dots, v_k = v$ s.t. $\forall_{i,j, i \neq j} v_i \neq v_j$ and $\{v_j, v_{j+1}\} \in E$

G undir is **CONNECTED** $\Leftrightarrow \forall u, v \in V \quad \exists$ path between u and v

CONNECTED COMPONENT is a maximal set of nodes s.t. every pair of nodes in the set is connected

• in Digraph a walk from u to v is a sequence $u = v_0, \dots, v_k = v$ s.t. $\forall_{i=0, \dots, k-1} (v_i, v_{i+1}) \in E$
 a **DIRECTED PATH** is a directed walk where no node is repeated

G directed is **STRONGLY CONNECTED** if $\forall u, v \in V, \exists$

a) directed path from u to $v \Rightarrow v$ is **REACHABLE** from u

b) " " from v to $u \quad u \leftarrow v$

both conditions hold since REACHABILITY is not symmetric

STRONGLY CONNECTED COMPONENT = maximal subset of vertices $C \subseteq V$ s.t. every pair of nodes in C is mutually reachable

given a digraph, the **UNDERLYING UNDIRECTED GRAPH** is obtained by ignoring the direction of the links in G .

$$(u, v) \Rightarrow \{u, v\}$$

a digraph is **WEAKLY CONNECTED** if its underlying undir graph is connected $\Rightarrow$ **WEAKLY CONNECTED COMPONENT** is the maximal subset of nodes that forms a connected component in the underlying undir graph

e.g. $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\pi: \begin{matrix} 1 \rightarrow 3 \\ 2 \rightarrow 2 \\ 3 \rightarrow 1 \end{matrix}$ $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

given an adj matrix A and a permutation matrix P , PA PERMUTE THE ROWS OF A

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \cdot A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \end{pmatrix}$$

while $A^T P^T$ PERMUTES THE COLUMNS of A

PERMUTES THE COLUMNS of A

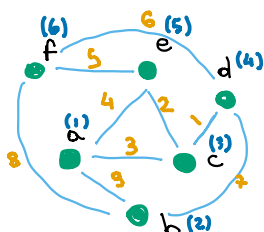
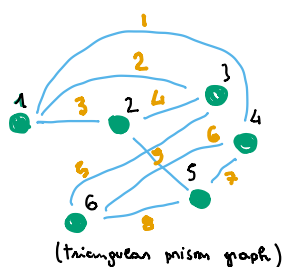
$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \cdot P^T = \begin{pmatrix} a_{13} & a_{12} & a_{11} \\ a_{23} & a_{22} & a_{21} \\ a_{33} & a_{32} & a_{31} \end{pmatrix}$$

$P A P^T$ PERMUTES ROWS and COLUMNS of A

GRAPH ISOMORPHISM TESTING $\rightarrow$ test whether two graphs G and H have the same structure up to a relabeling of the nodes

$$G = (V_G, E_G) \quad H = (V_H, E_H) \quad \text{s.t.} \quad |V_G| = |V_H| \quad \text{and} \quad |E_G| = |E_H|$$

H and G are isomorphic if $\exists$ bijection $\varphi: V_G \rightarrow V_H$ s.t. $\underbrace{(u, v) \in E_G \Leftrightarrow (\varphi(u), \varphi(v)) \in E_H}_{\text{adjacency is preserved}}$



$$\begin{array}{lll} \varphi(1) = c & \varphi(2) = a & \varphi(3) = e \\ \varphi(4) = d & \varphi(5) = b & \varphi(6) = f \end{array}$$

Let write the adjacency matrices for G and H

$A_G =$

0	1	1	1	0	0
1	0	1	0	1	0
1	1	0	0	0	1
1	0	0	0	1	1
0	1	0	1	0	1
0	0	1	1	1	0

$$A_H = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

if G and H isomorphic $\varphi: V_G \rightarrow V_H$

$$\begin{aligned} (u, v) \in E_G &\Leftrightarrow (\varphi(u), \varphi(v)) \in E_H \\ A_{G, u, v} &= A_{H, \varphi(u), \varphi(v)} \end{aligned}$$

→ define P s.t. $P_{\psi(u), u} = 1$

$$\Rightarrow P e_m = e_{\varphi(m)}$$

where $e_m = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}_m$

let's permute rows and columns of G

$$(\dot{P}A_G P^T)_{a,b} = (PA_G P^T)_{\varphi(u), \varphi(v)} = A_{G_{u,v}}$$

if $\exists$ permutation matrix P s.t.

$$P A_G P^T = A_H \Rightarrow G \text{ and } H \text{ are isomorphic}$$

$(PA_G P^T) \varphi(u), \varphi(v) = (A_H) \varphi(u), \varphi(v)$
 $(PA_G P^T) a, b = A_H a, b \quad \forall a, b \in V_H$
 $A_H = PA_G P^T$